

ZSWatch/ZSWatch

ZSWatch - the Open Source Zephyr™ based Smartwatch, including both HW and FW.

April 2026

Executive Summary

ZSWatch is a trailblazing open source smartwatch project, encompassing both hardware and firmware, built on the robust Zephyr RTOS and nRF Connect SDK. It offers unparalleled transparency, customization, and a comprehensive developer experience, addressing the market need for truly open wearable technology. With a feature-rich platform, extensive documentation, and a strong commitment to community-driven development, ZSWatch is positioned to become the premier platform for open hardware enthusiasts, embedded developers, and privacy-conscious users.

8.8/100

OVERALL SCORE

7.5

MARKET FIT

9.5

TECHNICAL

ZSWatch: The Open Source Smartwatch

Hardware and Firmware Built on Zephyr RTOS

- Enabling Transparency, Customization, and Innovation

Executive Speaker Notes:

Good morning. We're here to talk about ZSWatch, a revolutionary project in the wearable tech space. ZSWatch is not just another smartwatch; it's a complete open-source platform, from the circuits to the code, built on the powerful Zephyr RTOS. Our vision is to empower users and developers with unprecedented control over their devices.

The Closed Smartwatch Ecosystem

- Lack of Transparency: No insight into hardware or software.
- Limited Customization: Users cannot modify or extend functionality.
- Vendor Lock-in: Dependence on a single manufacturer's ecosystem.
- Stifled Innovation: Barriers to new features and community contributions.
- Privacy Concerns: Uncertainty about data handling in proprietary systems.

Executive Speaker Notes:

The current smartwatch market is a walled garden. Major players offer closed ecosystems where you can't see the code, can't change the hardware, and are locked into their services. This limits innovation, raises serious privacy concerns, and frustrates developers and power users who want more control. ZSWatch is here to change that paradigm.

ZSWatch: The Open Platform You Control

- ****Fully Open Source****: All hardware designs (KiCad) and firmware (GPL-3.0) available.
- ****Zephyr RTOS Power****: Enterprise-grade, secure, and modular real-time operating system.
- ****nRF Connect SDK****: Modern development environment with robust Nordic SoC support.
- ****Rich Sensor Suite****: IMU, pressure, magnetometer, light, RTC, mic, PMIC for versatile applications.
- ****Developer-Friendly UI****: LVGL v9 with XML-based design (LVGL Editor) and reactive subjects.
- ****Extensible App Framework****: Modular architecture for easy app creation and integration.
- ****WatchDK Development Kit****: Lowers entry barrier for developers and hobbyists.

Executive Speaker Notes:

Our solution is ZSWatch - a smartwatch built from the ground up to be open. We provide everything: complete hardware schematics, the entire firmware source code, and a powerful development kit. It's powered by Zephyr RTOS and Nordic's NCS, ensuring stability and performance. The UI is built with LVGL, making app development intuitive and efficient. This is about giving control back to the user and fostering true innovation.

Capturing the Open Hardware & Developer Market

- ****Niche, Passionate Market****: Targeting FOSS enthusiasts, DIY builders, and embedded developers.
- ****Educational Tool****: Ideal platform for teaching RTOS, embedded programming, and hardware design.
- ****B2B Opportunities****: Customization services for industrial wearables, specialized monitoring devices, or academic research.
- ****Growing Privacy Demand****: Appeals to users seeking transparent and auditable technology alternatives.
- ****Strong Community Potential****: Fostering an active developer community for sustained growth and contributions.

Executive Speaker Notes:

While we aren't chasing the mass consumer market directly, ZSWatch addresses a significant and growing demand in the open hardware and embedded development communities. This market is passionate, influential, and willing to invest in platforms that champion transparency and control. Furthermore, ZSWatch offers significant potential for B2B applications, acting as a flexible base for customized professional wearables. Its educational value is also immense.

Technical Excellence & Modern Stack

- ****Robust RTOS****: Zephyr Project and nRF Connect SDK on nRF5340 dual-core.
- ****Comprehensive Hardware****: Custom PCBs with diverse sensors (IMU, Barometric, Light, Mag, Mic, RTC).
- ****Advanced UI****: LVGL v9 with XML-based UI design via LVGL Editor, enabling rapid iteration.
- ****Seamless Connectivity****: BLE (ANCS, AMS, GadgetBridge, HID, AoA) with secure DFU via MCUboot/MCUMgr.
- ****Developer Workflow****: CMake/West build system, native simulator, extensive Pytest suite for HIL & power testing.
- ****Code Quality****: Strict coding conventions, extensive logging, and CI/CD ensure reliability.

Executive Speaker Notes:

Technically, ZSWatch is a powerhouse. We've chosen Zephyr for its enterprise-grade capabilities and the nRF5340 for its power efficiency. Our hardware is custom-designed, featuring a rich array of sensors, all documented in KiCad. The software is meticulously crafted in C, with a modular architecture and a user-friendly LVGL UI, supported by the visual LVGL Editor. Our development workflow is robust, including native simulation for fast iteration, extensive testing, and seamless firmware updates, all contributing to a high-quality product.

Sustainable Growth: Revenue & Community

- ****WatchDK Sales****: Primary revenue from development kit distribution (e.g., Elecrow).
- ****Customization Services****: Offering specialized development and integration for enterprise clients.
- ****Strategic Partnerships****: Collaborating with educational institutions, hardware companies, and tech organizations.
- ****Grants & Sponsorships****: Ongoing pursuit of funding from open-source and research initiatives.
- ****Community-Driven Enhancements****: Leveraging a vibrant developer community for feature contributions and bug fixes.
- ****Future Premium Offerings****: Exploring commercialization of production-ready, open-source smartwatch units.

Executive Speaker Notes:

Our business model is built on multiple pillars. We generate revenue through sales of our WatchDK development kits, which are accessible and enable widespread adoption. We also aim to offer customization and integration services for businesses needing bespoke wearable solutions. Continued support from grants, like NGIO, is vital. By fostering a strong community, we ensure continuous improvement and organic growth, with a long-term vision for offering production-ready open-source smartwatches to a broader market.

Unrivaed Advantages in Open Wearables

- ****Full Stack Openness****: The only truly open smartwatch spanning hardware and software, promoting transparency and trust.
- ****Superior Developer Experience****: Tools, documentation, and architecture tailored for easy development and customization.
- ****Robust & Secure Foundation****: Zephyr RTOS provides enterprise-grade reliability and security, a differentiator in open source.
- ****Modular & Extensible Hardware****: Development kits and extension boards simplify prototyping and innovation.
- ****Community Empowerment****: Fosters a vibrant ecosystem for education, collaboration, and collective innovation.
- ****Privacy by Design****: Users can audit every line of code, ensuring data privacy and control.

Executive Speaker Notes:

Our primary competitive edge is our unwavering commitment to open source. We're not just open in theory; we provide the KiCad files, the firmware, and a complete toolkit. This level of transparency is unparalleled. Our focus on developer experience means building on ZSWatch is a joy, not a battle. We're leveraging the security and stability of Zephyr, fostering a strong community, and delivering a platform where privacy and user control are paramount. This is a truly unique offering in the wearable market.

Addressing Key Challenges

- ****Market Niche****: Focus on targeted marketing to open hardware and developer communities.
- ****Hardware Scalability****: Strategic partnerships with manufacturers like PCBWay for efficient prototyping and potential mass production.
- ****Feature Evolution****: Prioritize core features and rely on community contributions for broader functionality.
- ****Funding Diversity****: Diversify funding beyond grants to include product sales and service contracts for long-term sustainability.
- ****Security & Maintenance****: Continuous integration of security best practices and a responsive update cycle, leveraging Zephyr's ecosystem.

Executive Speaker Notes:

Like any ambitious project, ZSWatch faces risks. The open-source market is smaller than the mass market, but we're focusing our efforts there. Scaling hardware production is a challenge, but our existing partnership with PCBWay for prototypes is a strong start. We manage feature evolution by prioritizing core functionality and empowering our community. To ensure long-term sustainability, we are actively diversifying our funding sources. Our commitment to robust testing and Zephyr's secure foundation helps mitigate security risks.

The Driving Force Behind ZSWatch

- ****Jakkra (Project Lead)****: Visionary founder, lead software architect, and hardware designer, with deep expertise in embedded systems and Zephyr RTOS.
- ****NLnet (Funding Partner)****: Financial support through NGIO Commons Fund, committed to open internet technologies.
- ****PCBWay (Hardware Partner)****: Sponsoring PCB manufacturing and assembly for prototypes.
- ****Growing Global Community****: Dedicated contributors across hardware, firmware, and documentation, fostering collaborative development.

Executive Speaker Notes:

ZSWatch is currently driven by a lean, dedicated team, spearheaded by Jakkra, the founder and lead architect, whose expertise in embedded systems and Zephyr is the cornerstone of this project. We're fortunate to have strategic partners like NLnet, providing critical funding, and PCBWay, supporting our hardware development. The project benefits immensely from a growing global community of developers and enthusiasts, contributing across all layers of the stack.

Investment in Open Innovation

- ****Seed Funding Secured****: Successfully obtained grants from NGIO Commons Fund via NLnet, validating project vision and impact.
- ****Hardware Sponsorship****: PCBWay's sponsorship significantly reduces prototyping costs.
- ****Lean Operations****: Focused development model minimizes burn rate.
- ****Future Capital Needs****: Scaling production, expanding software features, and enhancing market reach will require additional strategic investment.
- ****Value Proposition****: Investing in ZSWatch is investing in the future of open hardware and software, with potential for B2B licensing, custom solutions, and a strong community asset.

Executive Speaker Notes:

ZSWatch has demonstrated early financial viability with crucial grant funding from NGIO, validating its open-source mission. This, combined with hardware sponsorships, allows for very lean operations. While current funding supports development, scaling production, expanding advanced features like biometric integration, and broader market reach will necessitate additional strategic investments. We believe the unique value proposition of a fully open, customizable, and high-quality smartwatch platform offers attractive returns for investors aligned with open innovation.

Join the Open Smartwatch Revolution

ZSWatch: Your Watch, Your Code, Your Control.

- Visit zswatch.dev to learn more.
- Explore the GitHub repository.
- Order a WatchDK to start developing.
- Engage with our community on Discord.

Invest in a future where technology is open, transparent, and truly owned by its users.

Executive Speaker Notes:

ZSWatch is more than a device; it's a movement towards open, transparent technology. We invite you to join us on this journey. Visit our website, dive into our code, get a WatchDK to experience it firsthand, and connect with our passionate community. Together, we can build the future of wearables - a future that is open, customizable, and controlled by you. Thank you.